

Appendix

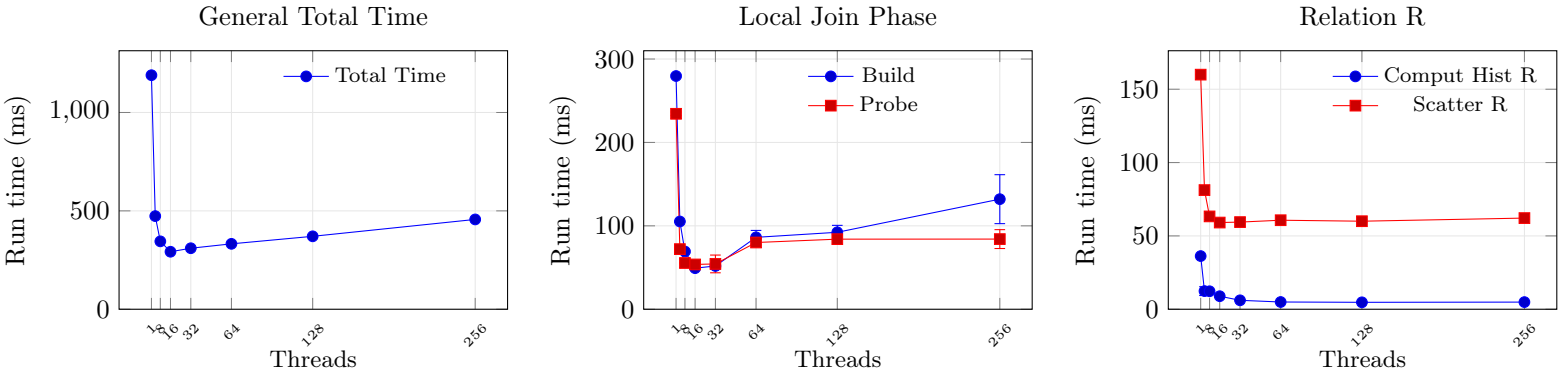


Figure 1: Strong Scaling (Uniform Data) Run time ms for Loop Level Solution

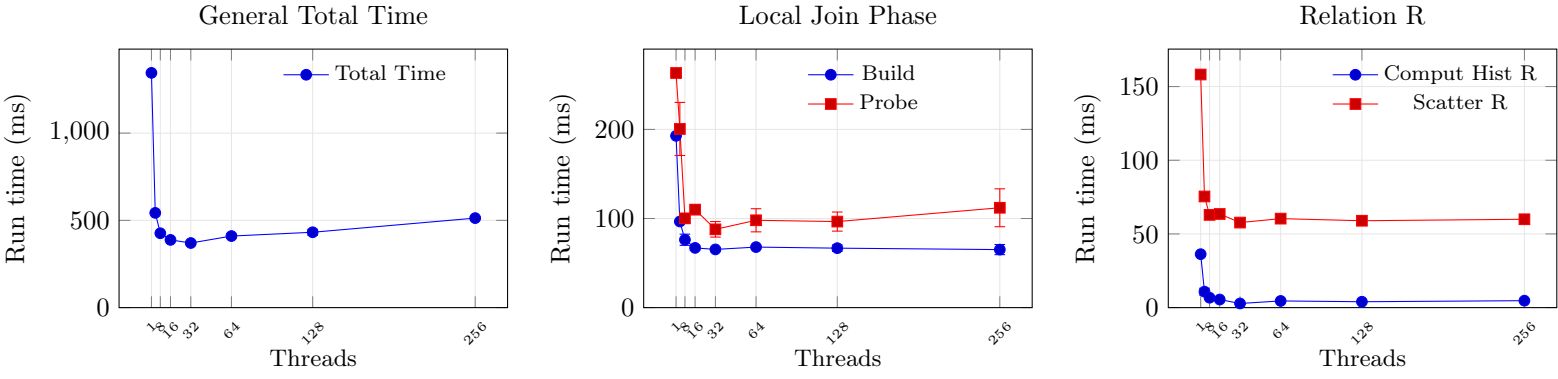


Figure 2: Strong Scaling (Skewed Data) Run time ms for Loop Level Solution

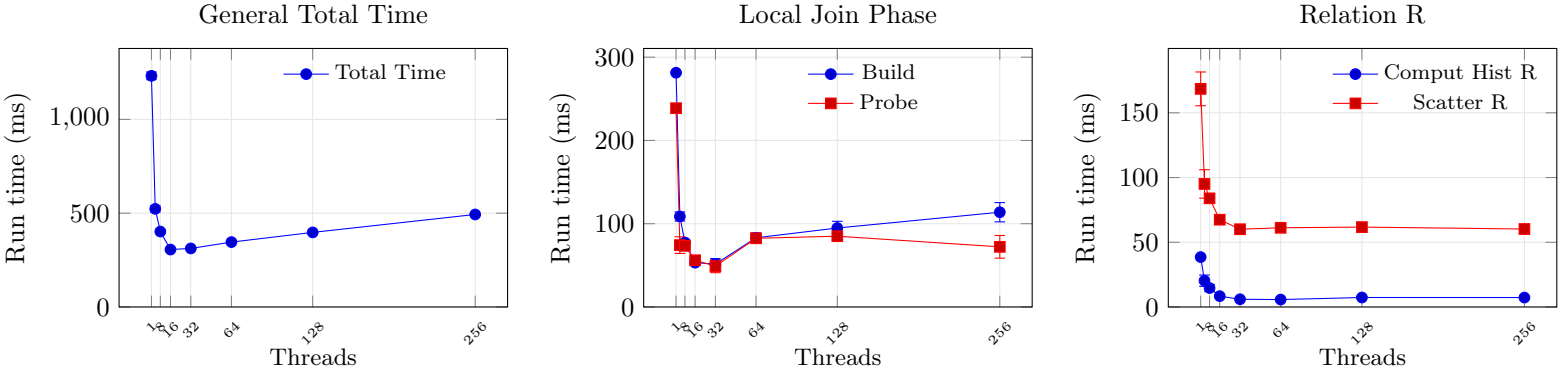


Figure 3: Strong Scaling (Uniform Data) Run time ms for Task Level Solution

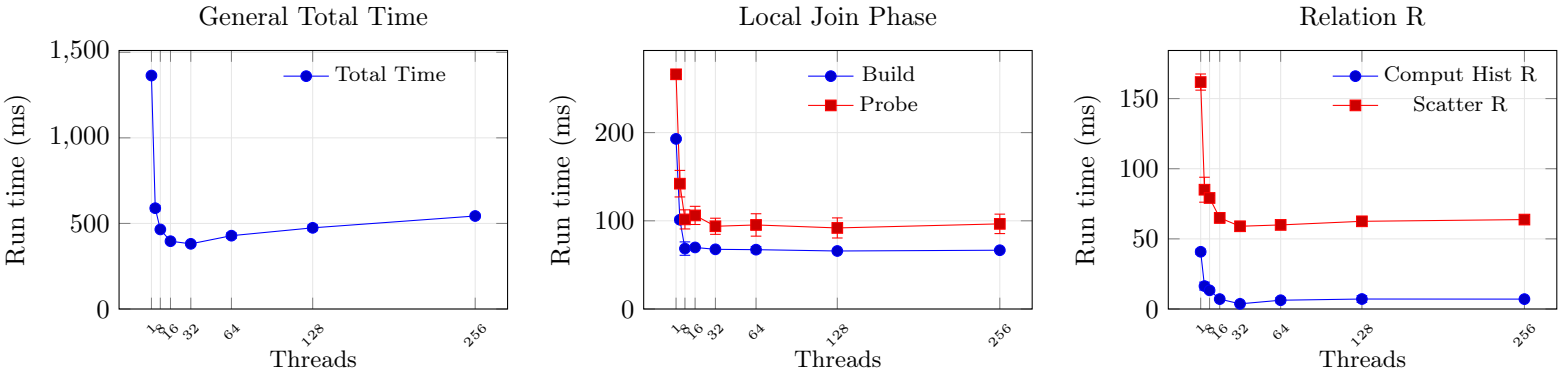


Figure 4: Strong Scaling (Skewed Data) Run time ms for Task Level Solution

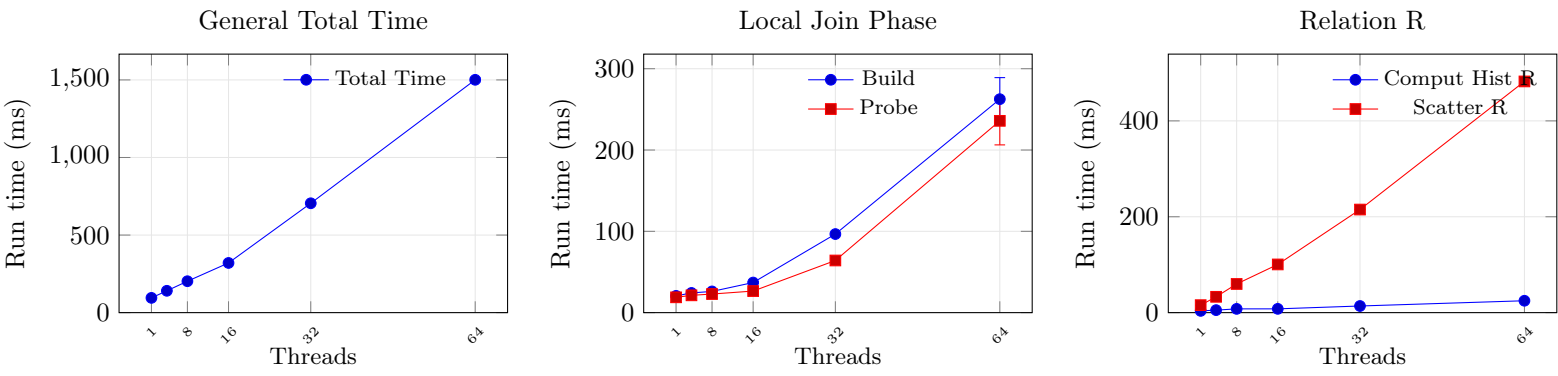


Figure 1: Weak Scaling (Uniform Data) Run time ms for Loop Level Solution

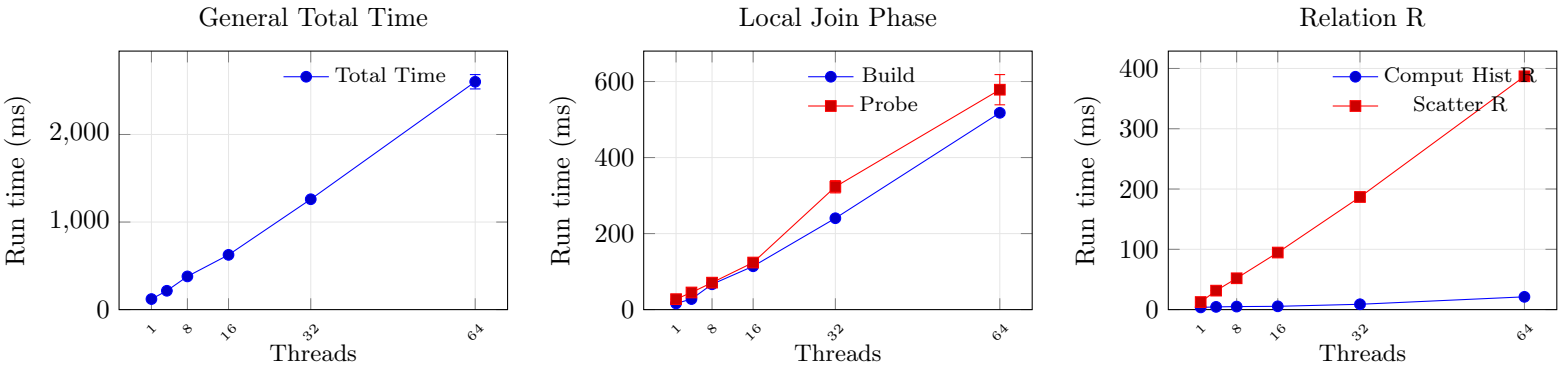


Figure 2: Weak Scaling (Skewed Data) Run time ms for Loop Level Solution

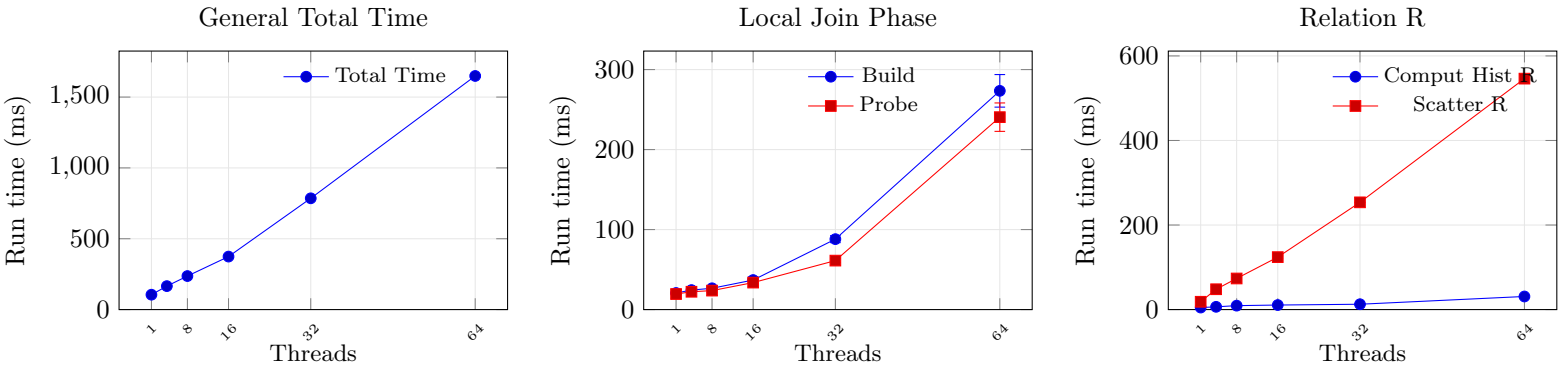


Figure 3: Weak Scaling (Uniform Data) Run time ms for Task Level Solution

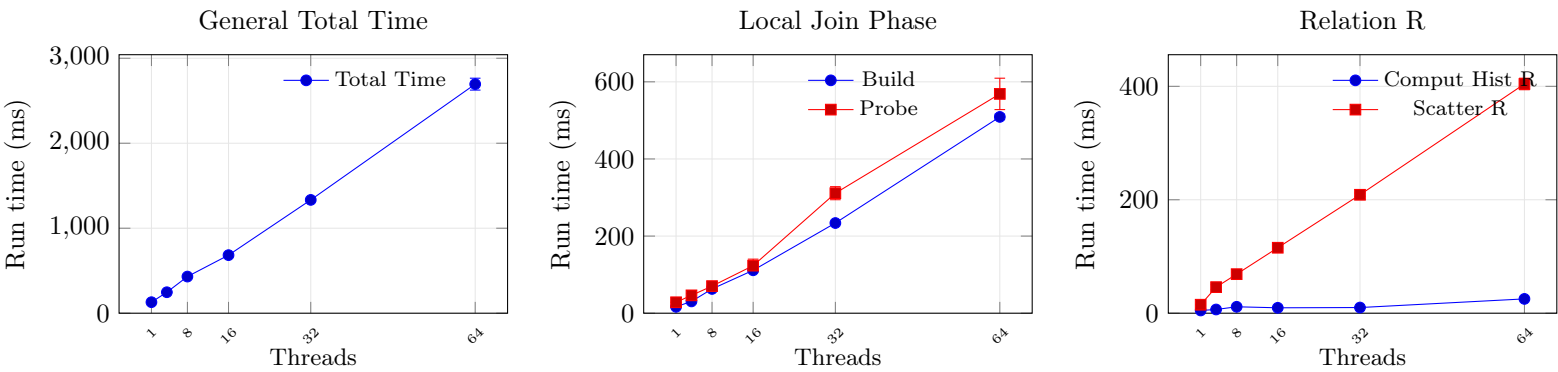


Figure 4: Weak Scaling (Skewed Data) Run time ms for Task Level Solution

Analysis of Buffer Optimization [MOD 2]

This is a small experiment I conceived afterward. In the version presented in Module 2, I had fixed the buffer size to 8; here, I investigate the effect of varying the buffer size. Therefore, I include it here as the first section of Module 3.

Table 1: Impact of Local Buffer Size on Scatter Phase and Relative Speedup Compared to Buffer Size 8

Buffer Size	Scatter R (ms $\pm$ std)	Speedup R	Scatter S (ms $\pm$ std)	Speedup S
8	67.38 $\pm$ 0.75	1.00 $\times$	66.19 $\pm$ 0.64	1.00 $\times$
16	62.66 $\pm$ 1.04	1.08 $\times$	63.30 $\pm$ 0.94	1.05 $\times$
32	60.33 $\pm$ 1.47	1.12 $\times$	59.89 $\pm$ 1.37	1.11 $\times$
64	59.05 $\pm$ 1.32	1.14$\times$	58.63 $\pm$ 1.62	1.13$\times$
128	60.56 $\pm$ 1.38	1.11 $\times$	60.09 $\pm$ 1.08	1.10 $\times$
256	64.01 $\pm$ 1.47	1.05 $\times$	63.55 $\pm$ 1.40	1.04 $\times$